

William Ward

wilward@umich.edu | (248) 880-5063 | linkedin.com/in/william-ward26 | github.com/billyward3 | williamward.dev | Ann Arbor, MI

PROFESSIONAL SUMMARY

University of Michigan CS graduate (Dec 2025) and technical co-founder. Solo-built a production mobile platform on GCP with 120 serverless endpoints, infrastructure-as-code with Terraform, and 1,849 automated tests. Designed a zero-knowledge encryption system inspired by Signal and Proton. Uses agentic AI development workflows (Claude Code + custom MCP servers) as a force multiplier for solo engineering.

EXPERIENCE

Roots App

March 2025 - Present

Technical Lead & Co-Founder | Ann Arbor, MI

- Architected a full-stack mobile platform (**Flutter/Dart, Node.js/TypeScript**) with 120 Cloud Functions on GCP, multi-bucket Cloud Storage, and signed-URL media pipelines. Incubated through the **LAUNCH Pre-Accelerator** and **Zell Lurie Institute** at the University of Michigan.
- Designed a **zero-knowledge encryption system** (X25519 ECDH, AES-256-GCM, HMAC-SHA256 blind indexing, Argon2id) with per-family key hierarchies and secure key rotation.
- Built **Terraform IaC** across dev/staging/prod, CI/CD via GitHub Actions, 1,849 automated tests, and two **custom MCP servers** for AI-assisted development with Claude Code.

IFE Americas

May 2023 - Aug 2024

Full Stack Developer / Database Developer | Brighton, MI

- Built an **automated email pipeline** (Python, Flask, SMTP, SQL Server) and an internal **lead-management tool** in SQL Server integrated with the company CRM.

PROJECTS

Crossword Creator

Next.js TypeScript Vercel Redis

Wordlist-first crossword generator with a **constraint-satisfaction engine**, AC-3 arc consistency, and three placement strategies. Adjacency-aware validation checks perpendicular runs against an English bigram table. Web Workers for parallel solving. Live at crossword.williamward.dev.

PufferPhish

React TypeScript AWS ML

ML-powered **Chrome extension** for real-time phishing detection. Combines a CodeBERT ONNX classifier with a rule-based engine (URL structure, domain reputation, page content). **Monorepo with 5 packages** deployed on AWS CDK. Senior capstone project.

CivicHousing

Next.js TypeScript Leaflet

Housing intake wizard scoring 76 real Detroit listings by household needs, with an interactive map and a **Top Trading Cycles** allocation simulation. Built in one day for the Google x CSG x T4SG 2026 Hackathon.

EDUCATION

B.S. Computer Science, University of Michigan

Aug 2022 - Dec 2025

GPA: 3.4/4.0 | Advanced Operating Systems, Computer Security, Cryptography, Networks, Computer Architecture, Web Systems, Data Structures & Algorithms, Machine Learning, Quantum Computing, Linear Algebra

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, C/C++, Dart, SQL, Java, Bash

Frameworks: React, Flutter, Next.js, Node.js, Express.js, Flask, Tailwind CSS

Cloud & Data: GCP, AWS, Terraform, Docker, GitHub Actions, Firestore, PostgreSQL, MongoDB, Redis

Security & AI: X25519 ECDH, AES-256-GCM, Argon2id, Claude Code, MCP server design, agentic workflows